

Advanced Statistics for Social Sciences 1

Statistical Thinking and Data Analysis with R

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Outline

Last class

- Conduct and interpret a hypothesis test (t-test)

Today's class

- Compute and interpret effect sizes
- Understand confidence intervals
- Bayesian statistics
- Bayesian estimation
- Bayes factor
- Modeling categorical relationships

What can statistics do for us?

DESCRIBE

- Summarize data
- Visualize patterns
- Estimate parameters

DECIDE

- Test hypotheses
- Compare groups
- Evaluate claims

PREDICT

- Build models
- Forecast outcomes
- Classify new data

The problem we are solving

We collected a sample. Now what?

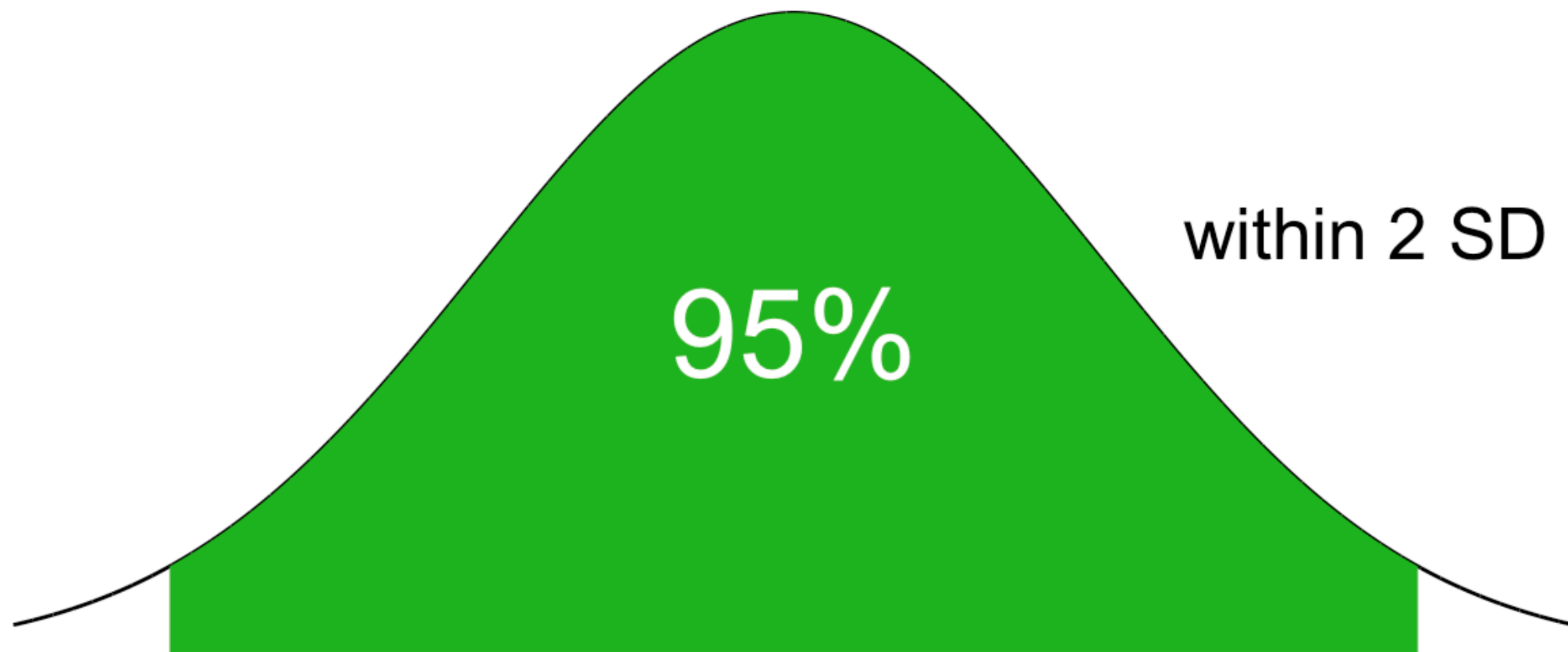
Three interconnected questions:

1. "Is this result statistically significant?" → HYPOTHESIS TESTING
2. "How big is the effect?" → EFFECT SIZES
3. "What range captures the parameter?" → CONFIDENCE INTERVALS

Key concepts to recap

The Normal Distribution and the ± 2 Rule

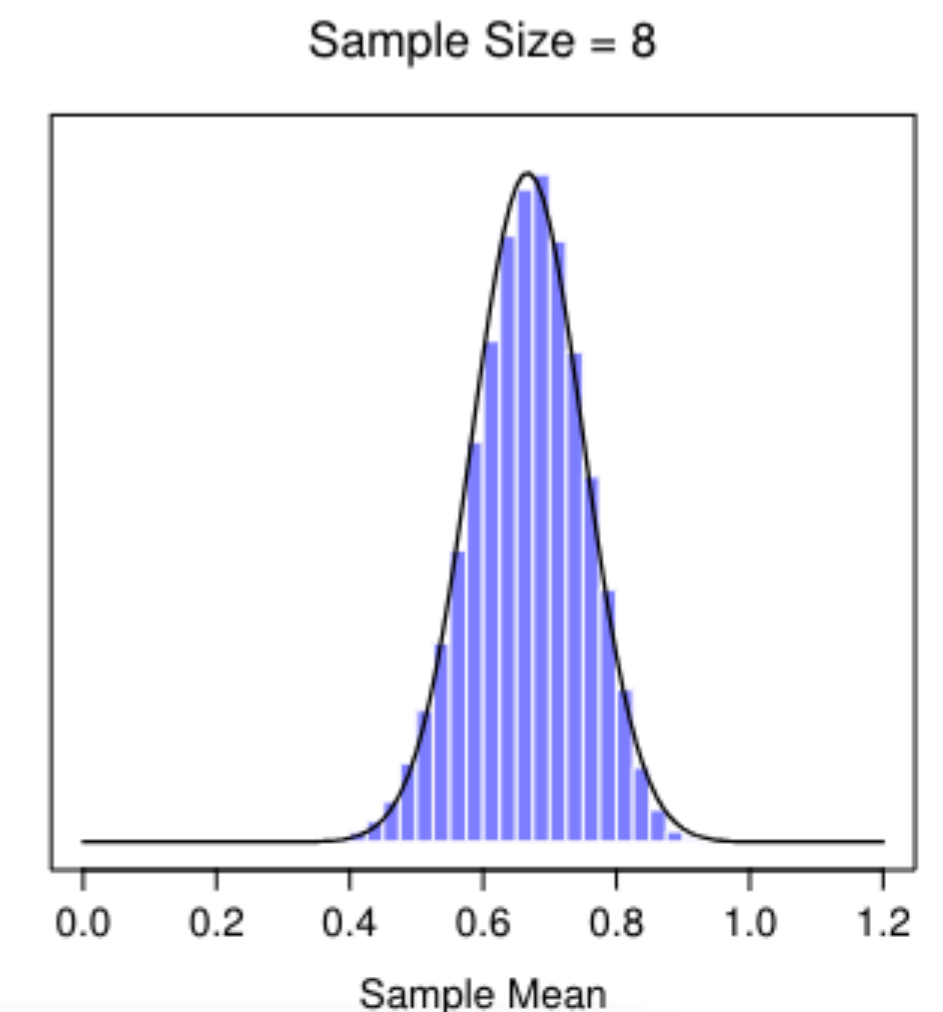
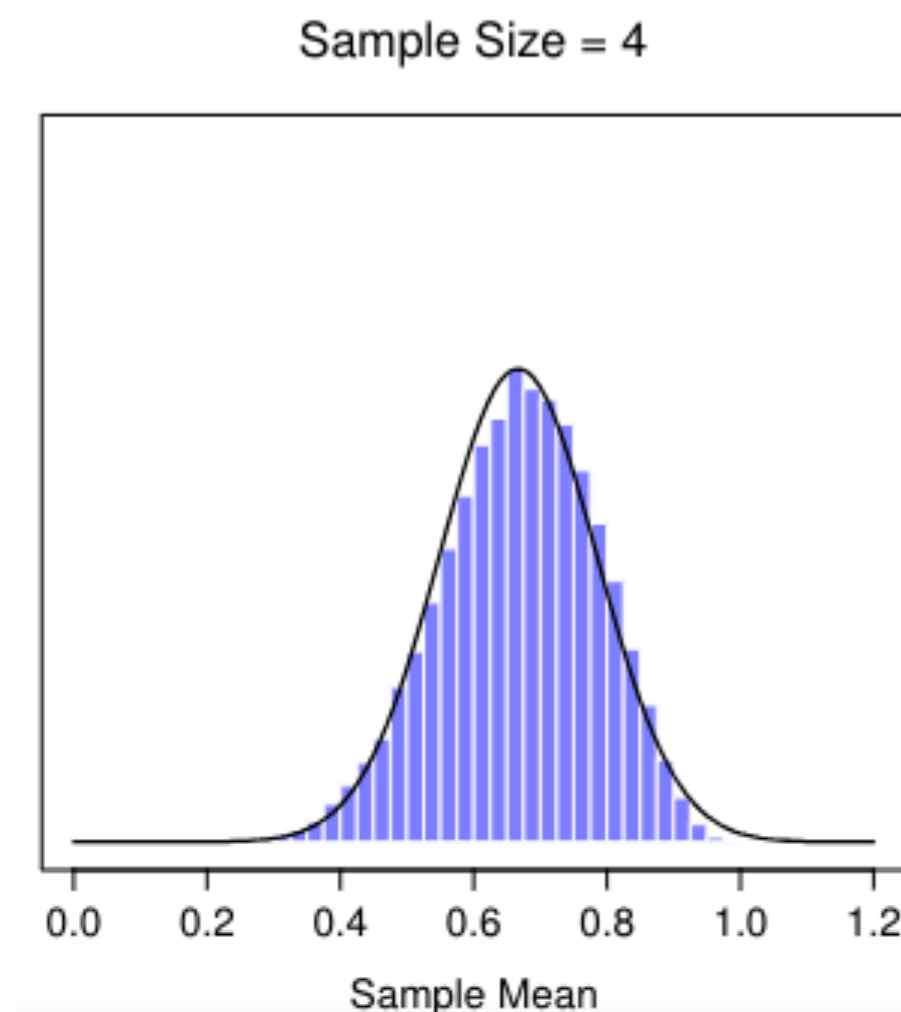
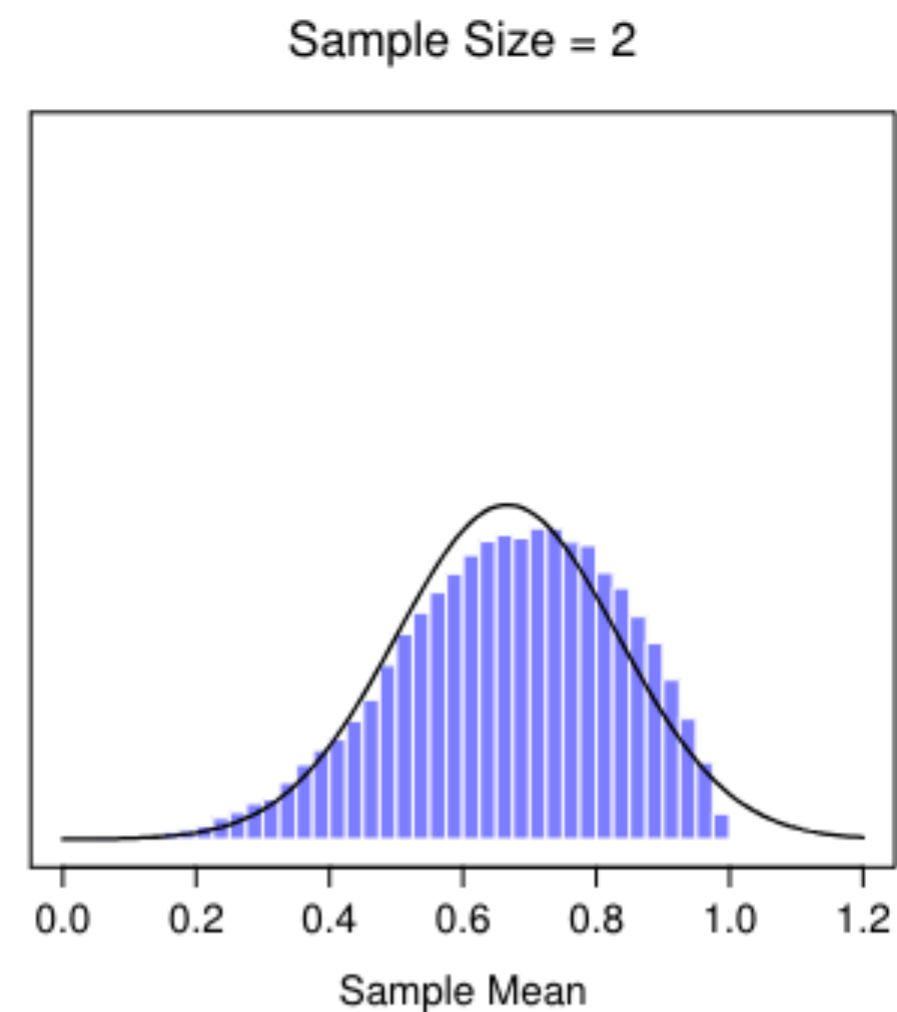
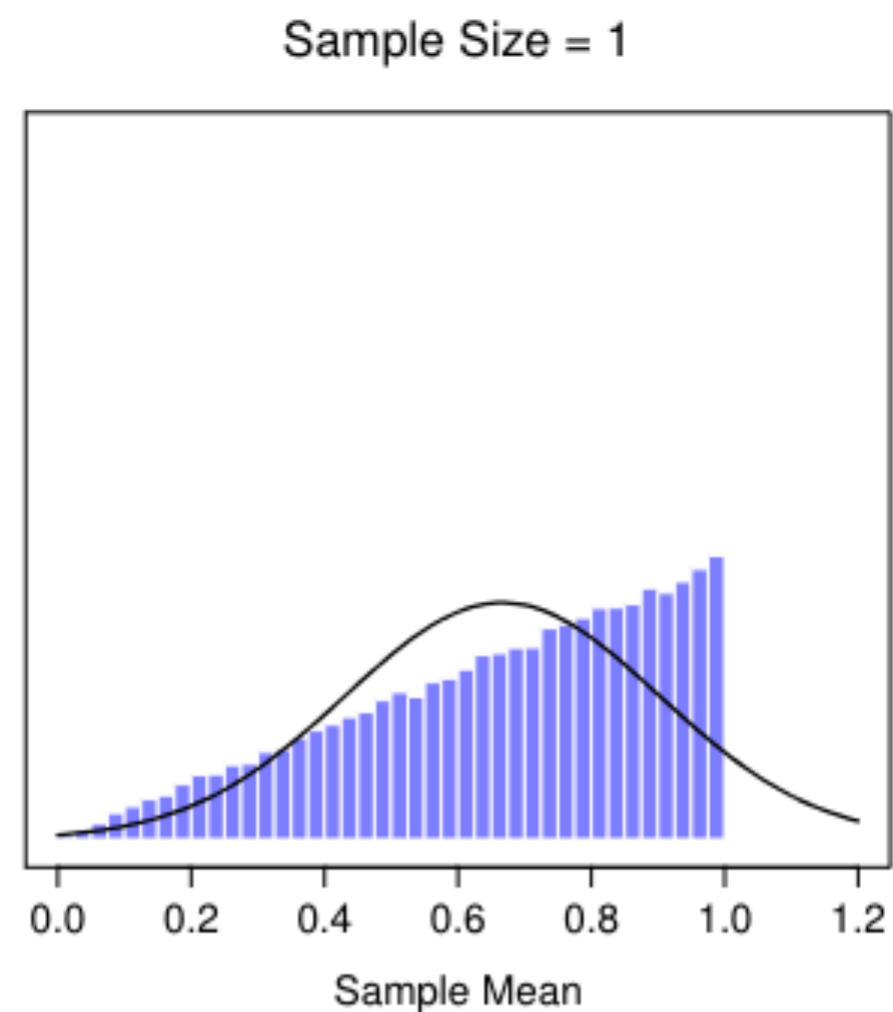
- About 95% of data fall within 2 standard errors of the mean
- This is important for both hypothesis testing AND confidence intervals



Key concepts to recap

Central Limit Theorem

- As long as your sample size isn't tiny, the sampling distribution of the mean will be approximately ***normal*** no matter what your population distribution looks like!



Quantifying effects

Confidence interval

Suppose that true population mean IQ is μ , standard deviation is σ

In your study, you collect N participants with mean IQ = $\bar{X}$

Recall that:

- Sampling distribution of the mean is approximately normal (central limit theorem)
- The sample mean is an unbiased estimator of the true mean
- 95% chance a normally-distributed quantity falls within two standard deviations of the true mean (actually 1.96 SD)

Confidence interval

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Mathematically, we can write that: $\mu - (1.96 \times \text{SEM}) \leq \bar{X} \leq \mu + (1.96 \times \text{SEM})$

Which can be rewritten as: $\bar{X} - (1.96 \times \text{SEM}) \leq \mu \leq \bar{X} + (1.96 \times \text{SEM})$

This range is the **95% confidence interval** (CI_{95})

- $\text{CI}_{95} = \bar{X} \pm (1.96 \times \sigma/\sqrt{N})$

Valid when N is sufficiently large for sampling distribution of mean to be normal

Confidence interval: common misconception

WRONG: "95% CI means 95% probability true value is in interval"

RIGHT: If you repeated the entire sampling and CI construction process, 95% of those resulting intervals would contain the true mean

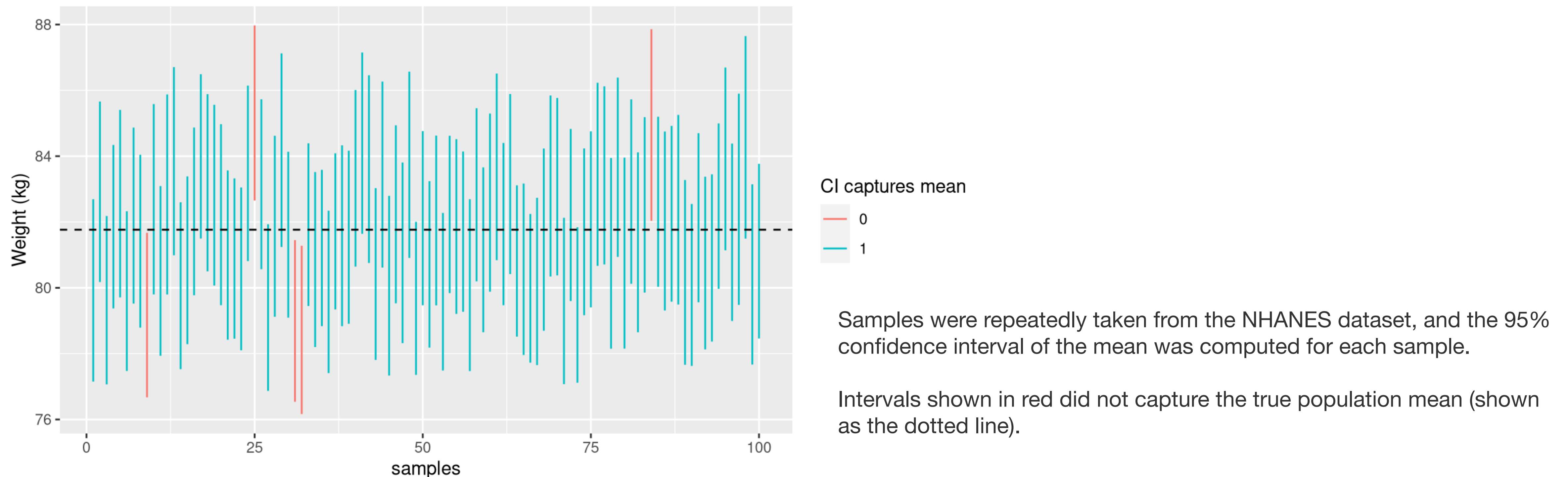
The "WRONG" interpretation treats the true parameter as random (it's not—it's fixed but unknown)

The correct interpretation treats the sampling procedure as random (if we could repeat the experiment, we would get different data and different intervals)

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RIGHT: If you repeated the entire sampling and CI construction process, 95% of those resulting intervals would contain the true mean



Confidence intervals and sample size

- Standard error decreases with sample size, so confidence intervals get narrower as N increases
- Larger samples provide tighter bounds on our estimate
- Diminishing returns: CI width is proportional to $1/\sqrt{N}$

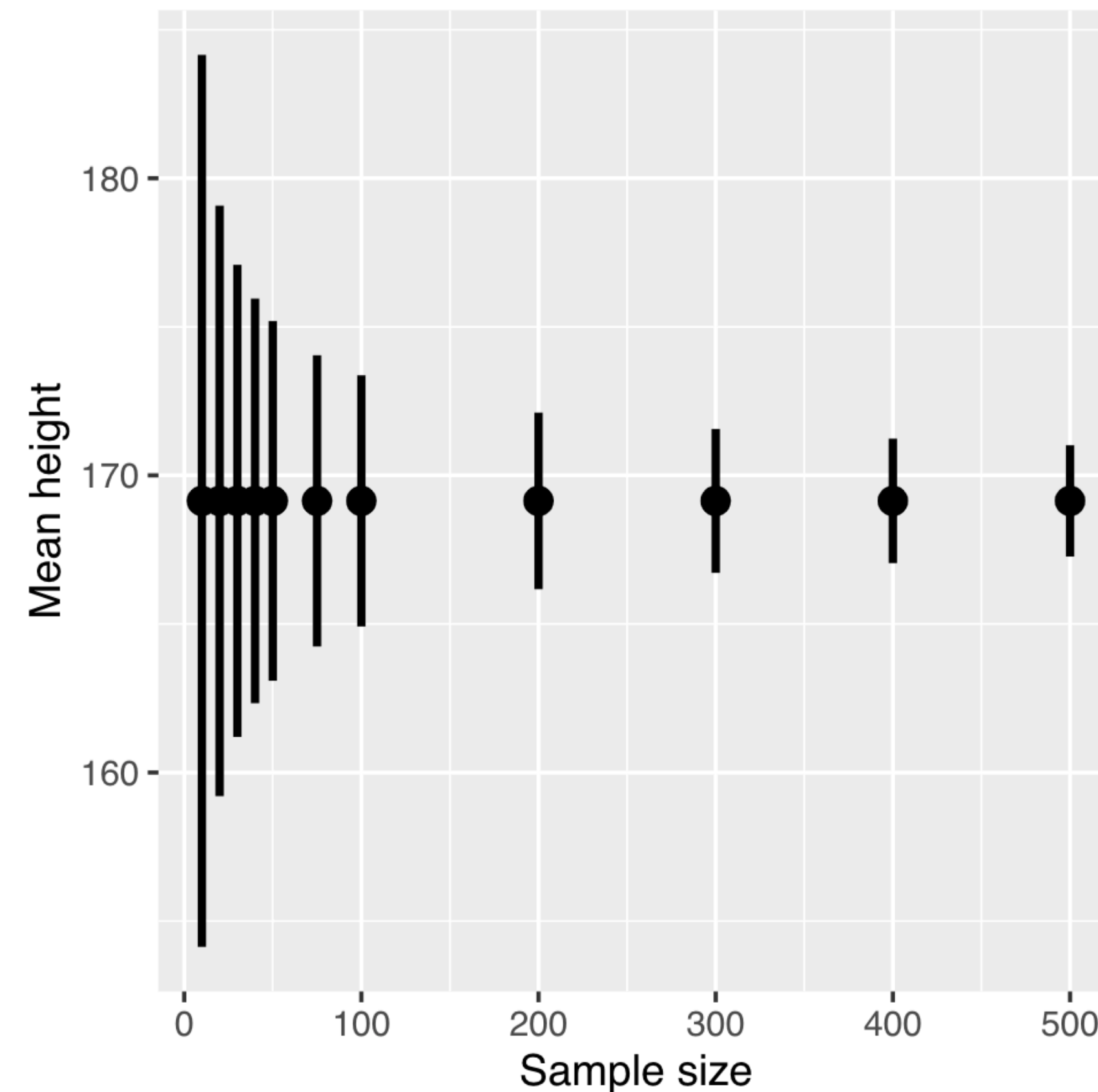


Figure 10.2: An example of the effect of sample size on the width of the confidence interval for the mean.

Effect sizes

“Statistical significance is the least interesting thing about the results. You should describe the results in terms of measures of magnitude – not just, does a treatment affect people, but how much does it affect them.”

- Statistical significance $\neq$ practical significance
- **Effect size:** Standardized measurement comparing the size of a statistical effect to a reference quantity (e.g., variability)
- Multiple ways to quantify effect size depending on the nature of the data

Cohen's D

- **Cohen's d:** Most common measure of effect size for comparing two means
- Formula: $d = (\bar{X}_1 - \bar{X}_2) / s$
- **s = pooled standard deviation:** Weighted combination of both groups' standard deviations
- $s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 + n_2 - 2)}}$

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- Key difference from t-statistic:
 - t-statistic denominator: standard error (includes sample size)
 - Cohen's d denominator: standard deviation (sample size “cancels out” asymptotically)
- **Result:** Cohen's d stays constant as sample size increases, while t-statistic grows
- Measures practical significance independent of sample size

Interpretation of Cohen's D

D	Interpretation
0.0 - 0.2	neglibible
0.2 - 0.5	small
0.5 - 0.8	medium
0.8 -	large

Effect sizes

1. Effect size measures magnitude on standardized scale
2. Not as dependent on sample size (unlike p)
3. Essential for assessing practical significance

Golden rule: Always report p -value along with effect size!

Three components of full inference — always include CI and effect size with p-values!

HYPOTHESIS TEST

Question: "Is null plausible?"

Answer: p-value (binary: reject or not)

Tells: "Is effect real?"

EFFECT SIZE

Question: "How big?"

Answer: d (on standardized scale)

Tells: "Is it practically meaningful?"

CONFIDENCE INTERVAL

Question: "What's the range?"

Answer: [lower, upper] bounds

Tells: "Direction, magnitude, precision"

Another common mistake to avoid

P-hacking (very, very common!!)

- running many tests, report only significant ones
- collecting data until $p < 0.05$ achieved (stop the count!)

Recommendation: Pre-register data collection plan, sample size, and hypotheses

Bayesian statistics

Recap: Bayes' Theorem

$$P(H|D) = \frac{P(D|H) * P(H)}{P(D)}$$

- *Prior $P(\text{Hypothesis})$* : Our degree of belief about hypothesis H before seeing the data D
- *Likelihood $P(\text{Data}|\text{Hypothesis})$* : How likely are the observed data D under hypothesis H?
- *Marginal likelihood $P(\text{Data})$* : How likely are the observed data, combining over all possible hypotheses?
- *Posterior $P(\text{Hypothesis}|\text{Data})$* : Our updated belief about hypothesis H, given the data D

Bayes theorem provides a way to understand data that can help solve many of the conceptual problems related to null hypothesis testing, while introducing new challenges

NHST focused on describing probability of data given a hypothesis (p-value)

$$P(H|D) = \frac{P(D|H) * P(H)}{P(D)}$$


- In our previous approach (frequentist approach), we did not believe in the idea of a probability of a hypothesis i.e., a hypothesis was either true or it wasn't.
- That is, ***the hypothesis was fixed and the data were random*** — hence, we focused on describing probability of data given a hypothesis (p-value)
- Bayesians are comfortable making probability statements about both data and hypotheses

Doing Bayesian Estimation

R notebook walkthrough

$$P(H|D) = \frac{P(D|H) * P(H)}{P(D)}$$

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Bayesian hypothesis testing

Having learned how to perform Bayesian estimation, let's try to use Bayesian methods for hypothesis testing..

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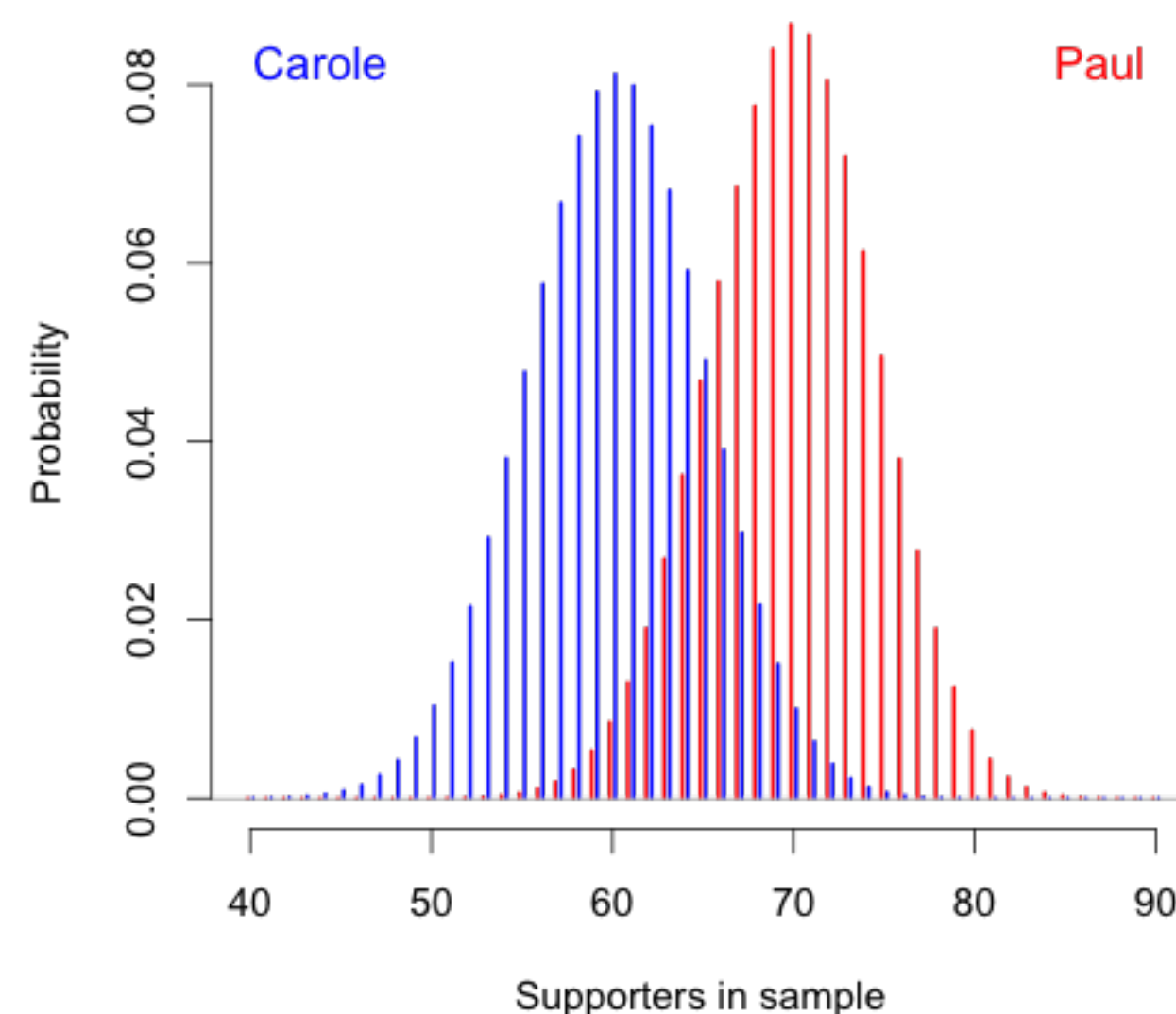
- Previously, we saw that NHST asks: “*How likely are these data if the null is true?*”
- When using Bayesian testing, we now ask: “*Which hypotheses better predicts the data we are observing?*”

This is a really big difference — we are now going to directly compare hypotheses, not just assume one is true!

Suppose that two researchers are interested in public opinion about public smoking bans. Paul believes that 70% of the public support such bans; Carole believes that the support is at 60%. Paul and Carole decide ask 100 randomly selected people whether they support public smoking bans.

Because Paul and Carole have very specific beliefs about the true support for the public smoking bans, they also have predictions about how the sample will turn out. Paul's best guess is that 70 out of 100 will support smoking bans, and likewise Carole's best guess is 60.

However, Paul and Carole can be more specific because the Binomial distribution is appropriate for model the kind of random sample. Figure below shows predictions of Carole (blue) and Paul (red).



Observed data: 62 people are supportive of the ban (out of the 100 people).

This seems to support Carole because the observation is closer to Carole's average prediction; however, Paul points out that he also predicted that 62 was possible, and so his hypothesis is not ruled out... how do we quantify which hypotheses better supports the data?

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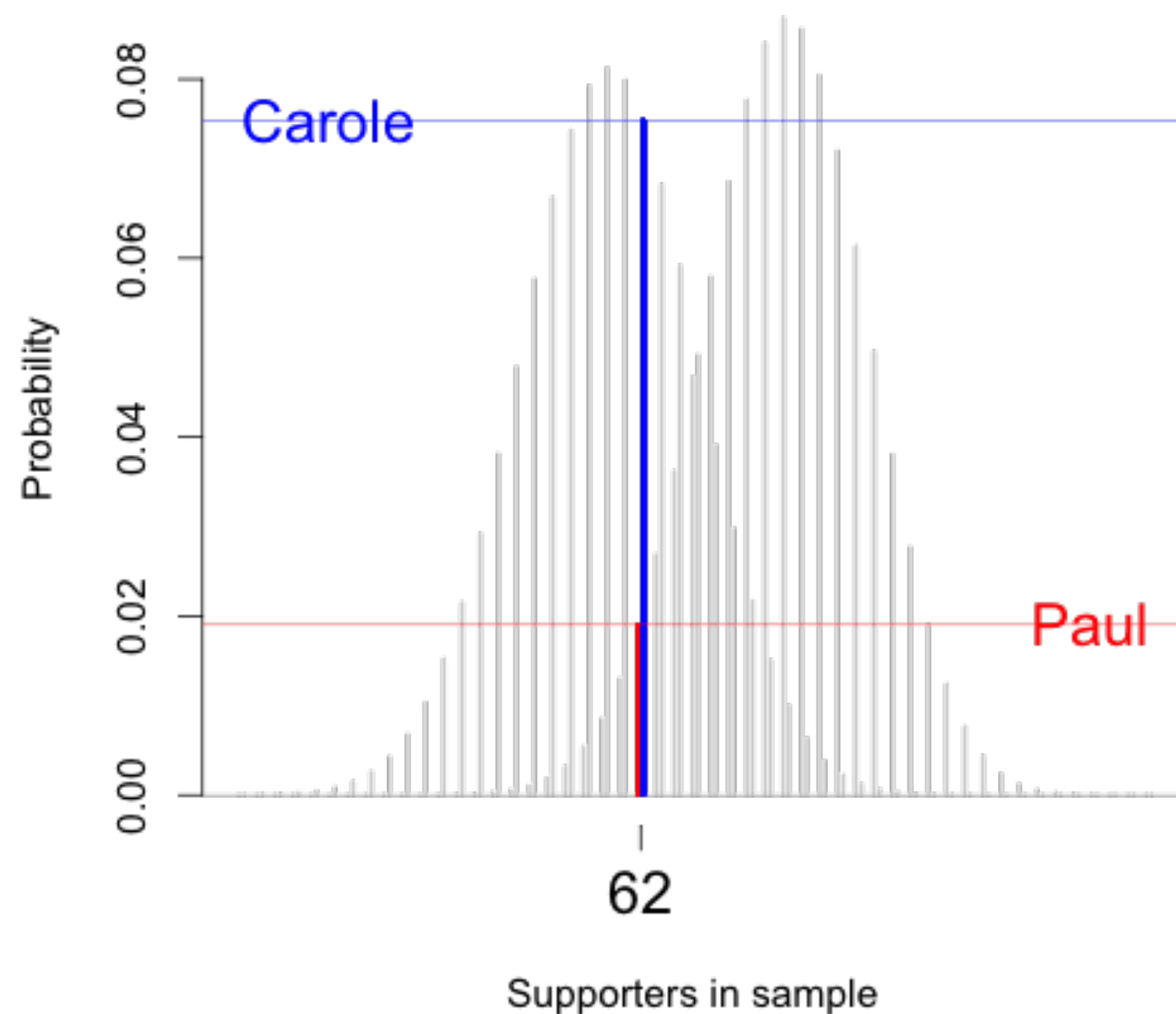
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Introducing the Bayes Factor

$$BF = \frac{p(data|H_1)}{p(data|H_2)}$$

The **Bayes factor** is the relative evidence in the data. The evidence in the data favors one hypothesis, relative to another, exactly to the degree that the hypothesis predicts the observed data better than the other.

- We can now easily compute the evidence in the data favoring Carole over Paul.
- Carole specified that the observation $y=62$ had a probability of 0.0754.
- Paul specified that the observation $y=62$ had a probability of 0.0191.
- Bayes factor tells us that under Carole's hypothesis, the observed data is much more likely than under Paul's.



$$\frac{\text{dbinom}(y, N, \text{carole})}{\text{dbinom}(y, N, \text{paul})} = 3.953$$

The observed data favors Carole by a factor of 4.

Bayes factors for statistical hypotheses

Previous example was rather easy:

- H_1 : $p = \text{exactly } 0.60$
- H_2 : $p = \text{exactly } 0.70$
- BF = simple ratio of two numbers

However, many of our examples are much harder:

- H_0 : null hypothesis i.e., difference = 0
- H_1 : difference is NOT 0 (but it could be 0.1, 0.5, 2.0, 100.0...?)

To get $P(\text{data} \mid H_1)$, we must:

- Consider all possible effect sizes, weight by prior probability and then average over all possibilities..
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However, Bayes Factor allows us to answer a much richer question..

We now have the ability to quantify the relative amount of evidence in favor of the null versus alternative hypotheses!

Fortunately: R handles the hard math with `ttestBF()` using default priors

Interpreting Bayes factors

Based on Kass & Raftery (1995)

BF	Strength of evidence
1 to 3	not worth more than a bare mention
3 to 20	positive
20 to 150	strong
>150	very strong

What They Answer

- **NHST:** "How likely are these data if H_0 is true?" (p-value)
- **BF:** "How much do the data favor H_1 over H_0 ?" (relative evidence)

Assumptions

- **NHST:** Assumes H_0 is true, asks if data are "extreme enough"
- **BF:** Compares both hypotheses on equal footing

What You Learn

- **NHST:** Binary decision (reject or fail to reject)
- **BF:** Continuous measure of evidence strength ($BF = 3, 10, 100\dots$)

Can You Support the Null?

- **NHST:** No! "Fail to reject" $\neq$ evidence FOR null
- **BF:** Yes! $BF < 1$ means data favor null over alternative

Interpretation

- **NHST:** $p = 0.02 \rightarrow$ "significant" (but how strong?)
- **BF:** $BF = 3.4 \rightarrow$ "positive but weak evidence" (quantified!)

Categorical data analysis

Modeling categorical relationships

We will now move toward **building models** that capture relationships in data.

We start with **categorical relationships**—relationships between variables measured as categories.

Categorical data are usually expressed in terms of counts; that is, for each value of the variable (or combination of values of multiple variables), how many observations take that value?

For example, when we count how many people from each major are in our class, we are fitting a categorical model to the data.

Goodness of Fit test (One variable)

Does a set of observed counts differs from some specific expected values that define the null hypothesis?

I purchased a bag of 100 candies, which are labeled as having 1/3 chocolates, 1/3 licorices, and 1/3 gumballs.

When I count the candies in the bag, we get the following numbers: 30 chocolates, 33 licorices, and 37 gumballs.

Question: Does the distribution of candy types match what the bag claims?

Chi-squared test:

Measures: How far are observed counts from expected?

$$\chi^2 = \sum_i \frac{(\text{observed}_i - \text{expected}_i)^2}{\text{expected}_i}$$

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Expected vs Observed:

Candy Type	Observed	Expected
Chocolate	30	33.33
Licorice	33	33.33
Gumball	37	33.33

$\chi^2 = 0.74$

The chi-squared statistic follows a known distribution (the chi-squared distribution) under the null hypothesis, allowing us to calculate p-values

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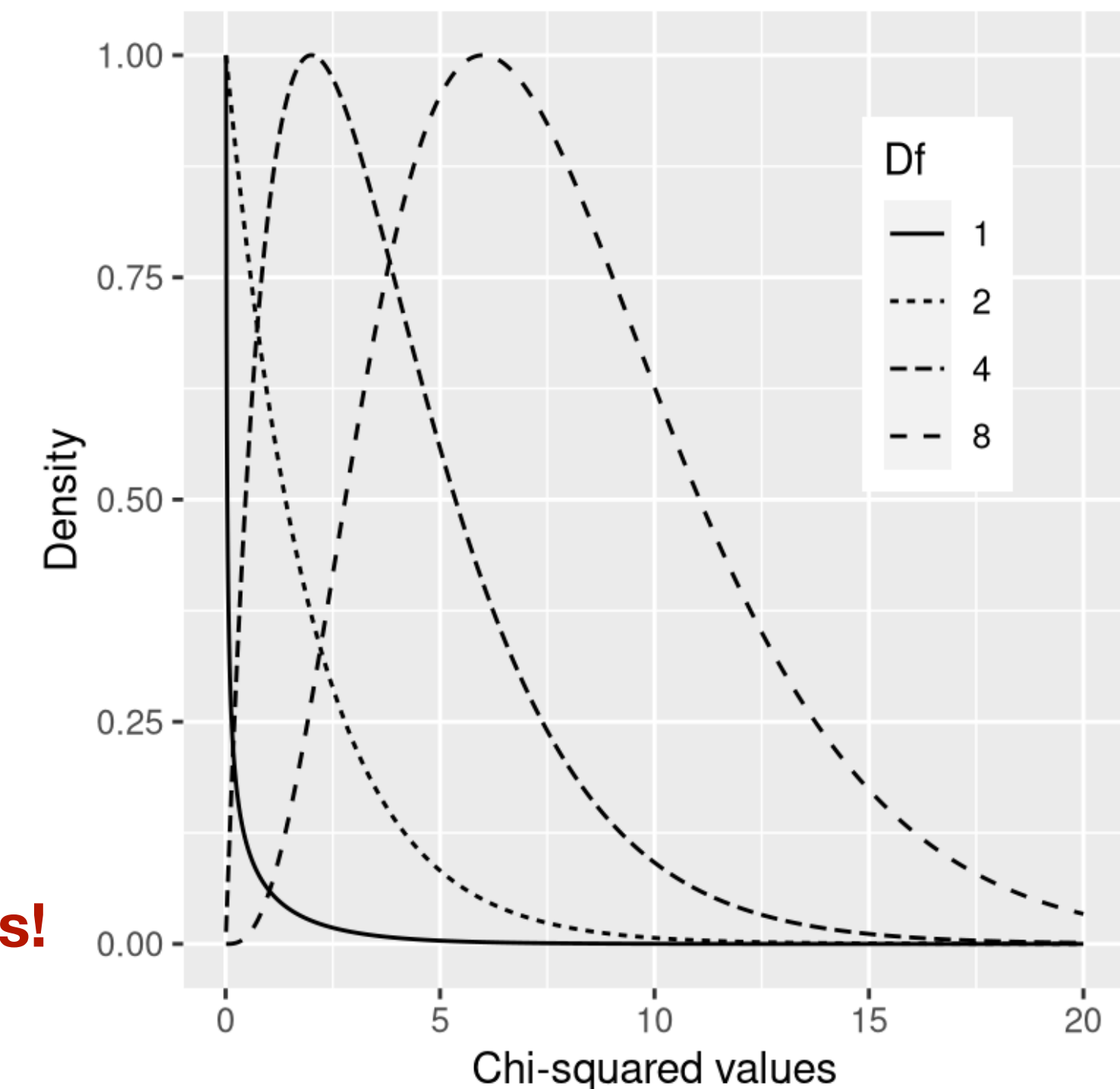
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$$\chi^2 = 0.74$$

$$P(\chi^2) > 0.74 = 0.691$$

Cannot reject the null hypothesis!



The chi-squared statistic follows a known distribution (the chi-squared distribution) under the null hypothesis, allowing us to calculate p-values

Summary: Chi-Squared test for one variable allows us to ask: “Does a variable match an expected distribution?”

Test of independence (Two variable)

Using Chi-Squared test, we can also ask: Are two variables related to each other?

Two variables: Race (Black/White) AND Search outcome (Searched/Not)

	Black	White
Searched	1,219	3,108
Not searched	36,244	239,241

Question: Is being searched related to driver race? Or are they independent?

Test of independence (Two variable)

	Black	White
Searched	1,219	3,108
Not searched	36,244	239,241

If independent: $P(\text{Searched AND Black}) = P(\text{Searched}) \times P(\text{Black})$

Calculate marginals:

- Overall search rate: $4,327/279,812 = 1.55\%$
- Black drivers: $37,463/279,812 = 13.4\%$

Expected if independent:

- Black + Searched: $279,812 \times 0.0155 \times 0.134 = \mathbf{585}$ expected
- But observed: **1,219** (way higher!)

Chi-squared test:

- $\chi^2 = 828$, $p \approx 0$
- **Conclusion:** Searches are NOT independent of race

Summary of today's class

- Compute and interpret effect sizes
- Understand confidence intervals
- Bayesian statistics
- Bayesian estimation
- Bayes factor
- Modeling categorical relationships